

massive spectrum, including the controlled introduction of near-degeneracies relevant for fine-cancellation mechanisms.

We illustrate the utility of this framework through several applications. First, we show that the standard clockwork mechanism emerges as a special case of an appropriate separable transformation. Second, we demonstrate that independent chiral zero-mode localization can be engineered even in symmetric theory-space Hamiltonians. Third, we analyze how Kaluza–Klein mass towers in deconstructed extra dimensions respond to such transformations, showing how the massive spectrum can be reshaped without altering the number of zero modes. Finally, we apply the framework to neutrino mass generation, where the transformations expand the phenomenologically viable parameter space and allow sub-eV masses to be obtained from TeV-scale physics within moderate lattice sizes, consistent with charged-lepton-flavor-violation constraints. It is important to emphasize the scope and intent of the present framework. The goal is not to engineer an arbitrary fermion mass matrix with prescribed eigenvalues and mixing matrices, which can always be achieved abstractly at the level of matrix parametrization by writing $M = U^\dagger D V$. Rather, we focus on identifying a restricted class of element-wise transformations that are compatible with theory-space constructions, preserve rank and nullity, and admit a clear physical interpretation in terms of site-local couplings. Within this constrained setting, the framework provides a systematic means of reshaping localization profiles and mass spectra while maintaining the underlying theory-space structure.

We emphasise that the framework is UV agnostic by construction; it characterizes the structural class of rank-preserving element-wise deformations available within theory space models, while the specific UV origin of any particular transformation is determined by the model under consideration. In concrete realizations such transformations can arise from physical parameters, for example, in deconstructed models, the mass matrix entries are set by site-dependent vacuum expectation values of scalar link fields, and different choices of these profiles lead to corresponding rescalings of matrix elements. Similarly, in extra-dimensional constructions, fermion masses arise from overlap integrals of left and right chiral wavefunctions, where the profiles of the two chiralities enter independently.

The remainder of this paper is organized as follows. Section II develops the theoretical framework for rank-preserving transformations, including proof sketches and physical interpretation, with complete mathematical details provided in the Appendix A. Section III presents applications to theory-space models: generalized clockwork constructions, independent chiral zero-mode engineering, controlled deformation of Kaluza–Klein spectra, and a worked example in neutrino mass generation. Section IV summarizes our results and discusses potential extensions.

II. RANK-PRESERVING SEPARABLE TRANSFORMATIONS OF MASS MATRICES

Theory-space constructions encode fermion masses in matrix structures whose null spaces correspond to massless modes in the four-dimensional effective theory. The localization profiles of these zero modes determine the resulting hierarchies in couplings and masses. In this section we introduce a class of index-dependent, element-wise transformations that preserve the number of zero modes while reshaping their profiles. This capability is the key to engineering controlled chiral structures in BSM model building.

A. Definition of the Transformation

Consider a fermion mass matrix A of dimension $m \times n$ arising from a theory-space construction. We define an index-dependent element-wise transformation producing a new matrix B with entries

$$b_{ij} = \frac{a_{ij}}{g_f(i, j)}, \quad (3)$$

where $g_f(i, j)$ is a nonvanishing function acting on the row and column indices. The label $f \in \mathbb{F}$ parameterizes a family of such transformations, with $g_f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{F}$ taking values in a field such as $\mathbb{R}$ or $\mathbb{C}$.

This operation may be viewed as a generalized Hadamard (Schur) scaling of the matrix entries [16]. However, unlike a uniform rescaling, it can in general alter the rank of A . Since massless fermions correspond to zero modes of the mass matrix, our interest lies in identifying the conditions under which the rank and nullity are preserved.

B. Rank and Nullity Preservation: The Separability Condition

The central structural property that governs rank preservation is the separability of the transformation function. We state the main result here and provide the detailed proof in Appendix A.

Main Result (Separability Condition). *The transformation $b_{ij} = a_{ij}/g_f(i, j)$ preserves the rank and nullity of A if and only if $g_f(i, j)$ is separable,*

$$g_f(i, j) = g_f^{(L)}(i) g_f^{(R)}(j). \quad (4)$$

To understand the origin of this condition, consider a row v_{i_0} of A that is linearly dependent on the remaining rows,

$$v_{i_0} = \sum_{j \neq i_0} \alpha_j v_j. \quad (5)$$